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DISCUSSION FORUM

# Principles for constructing notation in unit systems and their application to the SI

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**Abstract** The SI is the world's premier scientific instrument and, like all instruments, must be updated as science and technology advance. The CIPM is planning to redefine four SI base units in terms of invariants of nature and has already dubbed this revision the 'New SI'. This title will confuse most SI users, who are impacted only by the structure and notation of the SI, and neither of these has changed. A truly 'New SI', to be practical in the information age, would have at minimum an unambiguous, and preferably a machine- and user-friendly, notation. These and other principles for constructing a notation are proposed, and the SI historical and current notations are examined against these. Several levels of remediation are suggested.

**Keywords** SI units · Parsing · Notation · Ambiguity · Character set · Precedence

### Abbreviations

| | |
|---|---|
| BIPM | International Bureau for Weights and Measures |
| CIPM | International Committee on Weights and Measures |
| CCU | Consultative Committee on Units (of the CIPM) |
| CGPM | General Conference on Weights and Measures |
| ISO | International Organization for Standardization |
| ISQ | International System of Quantities |
| NIST | National Institute for Standards and Technology |
| NMI | National Metrology Institute |
| OASIS | Organization for the Advancement of Structured Information Standards |
| SI | International System of Units |

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## Introduction

The SI is the world's premier scientific instrument and, like all instruments, must be updated as science and technology advance. The CIPM is proposing [1] to redefine four SI base units in terms of invariants of nature and has already dubbed this revision the 'new SI' [2]. This title will confuse most SI users, who deal with only the structure and notation of the SI, and neither of these has changed.

The CIPM developed the SI in the 1950s, a time when digital computers were few and expensive and there was an assumption that they would only be used in large businesses (a computer designer reportedly [3] said in 1951, "all the calculations that would ever be needed (in the UK) could be done on three digital computers"). When the SI was declared in 1960 as a 'practical and evolving system', few anticipated that unit and quantity expressions might be widely written, stored, transmitted and read by machines as well as by humans. Today, unit expressions are universally written on a keyboard and rendered on a screen or through a printer; or generated by instruments and transferred across the laboratory bench, the internet or near space.

This paper examines the details of how SI unit and quantity expressions are written in text and in text-based data streams. A unit expression is a single unit symbol (e.g. kg, Pa, cd, Bq) or a combination of prefix, unit and operator symbols (e.g. µm, MΩ, kg/GJ and $m^3$ $kg^{-1}$ $s^{-2}$). A quantity expression is the product of a numerical value and a unit expression (e.g. 183 cm; $2.99 \times 10^{8}$ m/s; and $6.674\,28 \times 10^{-11}$ m$^{3}$ kg$^{-1}$ s$^{-2}$). Quantity expressions are ubiquitous in text (books, legislation, newspapers, web pages, technical and scientific documents); stand-alone unit expressions are mostly found in XML-based data streams.

## What is needed in the SI notation?

Price [4] has described principles for the metrological and definitional components of a quantity and unit system. An equally important component of a unit system is its notation, which is composed of symbols for units, prefixes and operators (precedence, number–unit product; unit–unit product and divisor). It should be noted that metric systems of units explicitly declare their notation as *symbols*, as distinct from the many different *abbreviations* used for, say, US customary units.

It is vital that SI expressions are written correctly to reduce the chance of misinterpretation by humans or machines. This is not simple, since for historical reasons, the SI notation is non-systematic, inconsistent and ambiguous [5]. This is further complicated by the different specifications for the SI published by BIPM and ISO. In particular, these authorities have independently changed several times the product symbols used in derived units, and the BIPM specification has some contradictions. Consequently, secondary SI sources can be wrong, and thus media and public SI usage is generally poor [6].

There is a clear desire among some scientists that the SI notation should be logical and consistent to reflect the logical structure of the SI and to make it easier to understand. Over the last 50 years, there has been much correspondence to journals [7–19], and discussions at the CCU [20, 21] regarding single-point changes to the prefix and unit symbols, typically:

- replace the two-character prefix da with a single character, such as D
- replace the lower-case prefix symbols h, k with uppercase H, K
- rename the base unit kilogram with a non-prefixed name and symbol, such as grave, giorgi, gali, symbol G
- make the kilomole a base unit and rename it without a prefix
- define new names and symbols for the unified atomic mass unit, impulse, electric field, loudness level, cycle, etc.
- replace the symbols using Greek characters ($\mu$ and $\Omega$) with Latin characters, such as u and Ohm
- define a single lowercase symbol for the litre, such as l, lt, ltr

How should the relative merits of the many suggestions be decided? It is suggested that scientists have wasted 50 years arguing for a systematic or consistent SI notation and that it is much more important to have a notation that is unambiguous and usable for the information age. Hence, six principles of quantity-unit notation are proposed here:

1. To enable unambiguous recognition by software, the symbols of a unit system should be referenced to a character in the Unicode character set [22].
2. To facilitate data transfer between software systems, the standard for a unit system should specify a representation for unit expressions in plain text.
3. To enable unambiguous representation of unit expressions, the symbols for base units and the combinations of prefix, unit and operator symbols for derived units should be syntactically unambiguous, and this should include the symbols for "units accepted for use with the system".
4. To enable unambiguous parsing of unit expressions, the standard for a unit system should state operator precedence.
5. To enable unambiguous parsing of quantity expressions, the standard for a unit system should have rules for writing a quantity expression, and the combinations of numbers, operators and unit symbols in quantity expressions should be syntactically unambiguous.
6. For convenience, symbols of a unit system should be easy to generate on a keyboard (both physical and virtual).

## What is found in the SI notation?

The notation in the BIPM and ISO standards for the SI is examined here against these principles. Editions of the BIPM 'SI Brochure' are abbreviated here as 'SI-1', 'SI-2', etc. ISO has published standards for the SI (ISO 1000) from 1969 to 1998 and for the ISQ (ISO 31) in 1981 and 1992. ISO combined its SI and ISQ standards in 2009 as ISO 80000 [23].

### The SI symbols do not have specified characters

The SI Brochures print the symbols used for units, prefixes and operators but do not specify any character information. It is assumed that the characters are drawn from the Latin and Greek alphabets and from mathematical symbols for the units degree, arcminute and arcsecond and the operators for precedence, multiplication, division and unary minus.

Lower and uppercase Latin characters are available in the ASCII, ISO-8859-1 (Latin 1) and Unicode character sets. Some candidate characters for the non-Latin letters are shown in Table 1.

**Table 1** Candidate SI characters and their hexadecimal codes in three character sets

| SI symbol | 7-bit ASCII | 8-bit ISO-8859-1 | Multi-byte Unicode |
|---|---|---|---|
| $\mu$ [micro] | × | MICRO SIGN (00B5) | GREEK SMALL MU (03BC) |
| Ω [ohm] | × | × | GREEK CAPITAL OMEGA (03A9); OHM SIGN (2126) |
| ° [degree] | × | DEGREE SIGN (00B0) | DEGREE SIGN (00B0) |
| ′[arcminute] | APOSTROPHE (0027) | APOSTROPHE (0027) | PRIME (2032) |
| ″[arcsecond] | QUOTATION MARK (0022) | QUOTATION MARK (0022) | DOUBLE PRIME (2033) |
| [unit product] | SPACE (0020) | SPACE (0020) | SPACE (0020) |
| · [unit product] | × | MIDDLE DOT (00B7) | DOT OPERATOR (22C5) |
| / [unit division] | SOLIDUS (002F) | SOLIDUS (002F) | DIVISION SLASH (2215) |
| × [number-product] | × | MULTIPLICATION SIGN (00D7) | MULTIPLICATION SIGN (00D7) |
| − [unary minus] | HYPHEN (002D) | HYPHEN (002D) | MINUS SIGN (2212) |
| ( ) [precedence] | PARENTHESIS (0028, 0029) | PARENTHESIS (0028, 0029) | PARENTHESIS (0028, 0029) |

ISO published a standard ISO 2955 [24] for representing SI unit symbols using the (127-character) ASCII character set, which represented '°' as 'deg', '°C' as 'Cel', 'Ω' as 'Ohm' and '$\mu$' as 'u'. This was withdrawn in 2001 after the publication of the much larger Unicode character set that contains all the SI symbols.

The typesetting of quantity and unit symbols is specified in SI-8 as: units are written in Roman font, while quantities are written in italic font. Dimension symbols are shown without comment as bold Roman font. The width of spaces is unspecified.

ISO 80000 has identical specifications for typesetting quantity and unit symbols, but shows dimension symbols as un-bolded Roman font.

## The SI does not specify a plain-text format for unit expressions

The SI notation was developed for typesetting unit and quantity expressions for human readability and requires the use of subscript and superscript formatting capability, as well as line-shifting and underlining, to render unit expressions such as '$\mathrm{m{\cdot}s^{-1}}$' or '$\frac{\mathrm{m}}{\mathrm{s}}$'. Much data transfer, program source code and email are presented in plain text, a stream of symbols with no formatting content, which cannot represent the SI mathematical structures. 'Plain text' was used to mean 'ASCII text' containing only the 95 printable ASCII characters, but it now applies to a stream of printable Unicode characters.

ISO 2955, although nominally about character substitution, also contained rules for representing SI unit expressions in plain text, which is about typesetting substitution. It specified the product symbol as the period (.), division symbol as solidus (/) and exponentiation as a trailing number, e.g. 'kN.m-2' or 'kN/m2' for the typeset '$\mathrm{kN/m^2}$'.

## Some SI unit expressions are ambiguous

### *Syntactic ambiguity from the implicit product symbol*

While it would be easier for humans if the SI symbols were unique, this is historically not the case and there are two symbols that represent both a prefix and unit, 'm' (milli, metre) and 'T' (tera, tesla). Where the product symbol is implicit (as per the style of a quantity equation, e.g. $E = mc^2$), it is possible to have syntactically ambiguous combinations of symbols, e.g. 'mN'. The BIPM and ISO have been aware of this latter problem since 1960, but have not addressed it very effectively.

- SI-1 and SI-2 [25, 26] stated that "The product of two or more units is preferably indicated by a dot. The dot may be dispensed with when there is no risk of confusion with another unit symbol, *for example* N·m *or* Nm *but not:* mN". (Note that the sentence is about the product of *units*, so it should have stated "…when there is no risk of confusion with a prefix symbol".)
- SI-3–SI-5 [27–29] added the period as an optional product symbol but evidently decided that the implicit product rule was too confusing, and stated "the product of two or more units may be indicated in any of the following ways, *for example* N·m, N.m, or N m".
- SI-6 [30] withdrew the period as a product symbol and stated "the product of two or more units may be indicated in either of the following ways, *for example* N·m or N m".
- SI-7 [31] stated "half-high dots or spaces are used to express a derived unit formed from two or more other units by multiplication. *Example* N · m or N m"
- SI-8 [32] states "In forming products and quotients of unit symbols the normal rules of algebraic multiplication or division apply. Multiplication must be indicated

by a space or half-high (centred) dot (·), since otherwise some prefixes could be misinterpreted as a unit symbol", with a marginal example shown as "N m or N · m".

- All editions of ISO 1000 [33–37] stated: "When a compound unit is formed by multiplication of two or more units, this should be indicated in one of the following ways: N · m, N m NOTES 2 In systems with limited character sets a dot on the line is used instead of a half-high dot. 3 The latter form may also be written without a space, provided that special care is taken when the symbol for one of the units is the same as the symbol for a prefix, e.g. mN is used only for millinewton, not for metre newton". (It is not obvious that "provided that special care is taken" means 'except', and that not "one of", but only the leading unit symbol needs to be unambiguous. The SI has only 'base' and 'derived' units; the text would be clearer if 'compound' was replaced by 'derived').
- ISO 80000-1 withdrew the period as a product symbol and states: "A compound unit formed by multiplication of two or more units shall be indicated in one of the following ways: N · m, N m. NOTE The latter form may also be written without a space, i.e. Nm, provided that special care is taken when the symbol for one of the units is the same as the symbol for a prefix. This is the case for m, metre and milli, and for T, tesla and tera". (Similar remarks apply).

In summary:

- The specified product symbols have changed several times over 50 years and are still different between the BIPM and ISO specifications.
- None of the ISO and BIPM editions of the SI address the duplicate prefix-unit symbol ambiguities that apply to the 'non-SI units accepted for use with the SI'. Thus the symbols 'h' (hecto, hour) and 'd' (deci, day) have the same ambiguities identified above. Does ISO 80000 allow the ambiguous derived units 'dlm' (day-lumen or decilumen) and 'hW' (hour-watt or hectowatt)?

*Contradictory typography of the product symbol*

- In SI-7 and SI-8, the text states that the product symbol is a half-high dot or space, but contradictorily prints the product symbol as "N · m" (i.e. space-dot-space), and also repeats this in its Table 3 'SI derived units with special names and symbols'.
- Similarly, ISO 1000 and ISO 80000 print the product symbol as "N · m" (i.e. space-dot-space), which contradicts the text of SI-8.

*Other ambiguous combinations of symbols*

Historically and currently in the SI Brochures, there are other ambiguous combinations of symbols, some arising from 'units which are accepted for use with the SI'.

- For all versions of the SI, the prefix symbol 'da' contains two prefix symbols, 'd' and 'a', in contravention to the semantic rule that forbids the concatenation of prefixes. (Note that a unit symbol such as 'ha' cannot be syntactically parsed as two prefixes, because it will have a product symbol separating it from other units, e.g. 'ha·m' as a metric analogue of the US customary unit of volume 'acre-ft').
- In SI-1 and SI-2, 'lm' (lumen or litre-metre), 'rad' (rad and radian), 'atm' (attotonne-metre or atmosphere).
- In SI-1–SI-7, 'da' (deciare or deca).
- In SI-6–SI-7, 'Pa' (petaare or pascal).
- In SI-6–SI-8, 'cd' (candela or centiday).

## The SI does not specify precedence for unit expressions

In mathematical expressions, operators are parsed left-to-right in this order of precedence [38]:

1. Parentheses
2. Factorial
3. Exponentiation
4. Multiplication and division
5. Addition and subtraction

The SI does not specify precedence for unit expressions, but precedence of prefixes is stated in SI-8 (section 3.1) as "The grouping formed by a prefix symbol attached to a unit symbol constitutes a new inseparable unit symbol (forming a multiple or submultiple of the unit concerned) that can be raised to a positive or negative power and that can be combined with other unit symbols to form compound unit symbols".

## The SI Brochure now specifies how to write a quantity expression

SI-1–SI-7 made no statement of how to write a quantity expression.

SI-8 added a new section 5.3.3 'Formatting the value of a quantity', stating: "The numerical value always precedes the unit, and a space is always used to separate the unit from the number. Thus the value of the quantity is the product of the number and the unit, the space being regarded as a multiplication sign (just as a space between units implies multiplication)".

ISO 1000 stated the rule for writing a quantity expression in section 6 'Rules for writing unit symbols': "…the unit symbol shall be placed after the complete numerical value in the expression for a quantity, leaving a space between the numerical value and unit symbol".

ISO 80000 section 6.5.6 'Other units' contains an orphan sentence "To express values of physical quantities, Arabic numerals followed by the international symbol for the unit shall be used". A new section 7.1.4 'Expressions for quantities' states: "The symbol of the unit shall be placed after the numerical value in the expression for a quantity, leaving a space between the numerical value and the unit symbol".

### Some SI symbols are not available on a keyboard

Few of the symbols in Table 1 are represented on US or Western-European keyboards, since these were developed to support the ASCII character set, nor on the virtual keyboards on mobile devices.

## Consequences of the ambiguities in the SI notation and specifications

### Unspecified character set

The SI's unspecified character set is perfectly suitable for print-based communication, but not for digital communications, where software has to map characters to SI symbols. For example, should software interpret a keyboard 'solidus' (/) as the symbol for unit division, or should this be mapped to the mathematical operator 'division slash' (/)? The absence of a standard has led to computer programmers and users choosing their own mappings, with consequent incompatibilities.

### Plain-text representation of unit expressions

The unit-aware calculators Google Calculator [39] and Wolfram|Alpha [40] do not recognise the ISO 2955 expression '3 kN.m-2', which is understandable as this standard is withdrawn. In the absence of such a standard, these systems, as well as the example shown below and many others, have developed their own conventions for writing unit expressions in plain text. This is obviously undesirable for system interoperability.

### Ambiguous unit expressions

#### *General*

The changing product symbol has led to confusion and poor SI usage in the press, public documents and signage. A common error is the ongoing use of the implicit product, e.g. 'kWh'.

The SI brochure visual contradictions about the product symbol lead to misinterpretations in secondary SI sources (e.g. [41]), which states (p. 179) "where units are formed by the multiplication of other units, this can be shown by using a raised dot with a space either side, or simply a space".

#### *Unit ambiguity in data streams*

The loss of the NASA spacecraft Mars Climate Orbiter was attributed to "Failure to use metric units in the coding of a ground software file, 'Small Forces', used in trajectory models" [42]. The thrust required for trajectory corrections was calculated by the ground navigational software in pound-seconds, but was applied by the spacecraft control software to the thrusters in newton-seconds, less than a quarter of the required thrust. Most commentators [43–45] drew this lesson: don't mix units!

There is, however, a more important lesson to be learned. This incident was an extreme example of unit ambiguity in a data stream: the ground software transmitted a *pure number* to the spacecraft software, which incorrectly interpreted it as a SI unit. When transmitting the value of a physical quantity, it is clearly vital to include both the number and unit, so that the receiving computer can reject or convert the unit.

Recent standards for data transfer are usually developed in XML for a particular scientific domain (ISO, WDTF, OGC) [46–48]. Each standard is faced with the problem of encoding units and quantities as text. One solution is to adopt a limited repertoire of quantity names and unit symbols to suit the particular domain. In the abbreviated code snippet below, the quantity is identified and the 'uom' field specifies 'units of measurement', using SI-like unit symbols:

```
⟨wdtf : Measurement⟩
  ⟨om : samplingTime⟩2011 − 08 − 20T16 : 12 : 01 + 10 : 00⟨/om : samplingTime⟩
  ⟨om : observedProperty title = "Electrical Conductivity @ 25C''/⟩
  ⟨wdtf : result uom = ''uS/cm''⟩90.00⟨/wdtf : result⟩
⟨/wdtf : Measurement⟩
```

Other data transfer standards [48] adopt unit symbols from ISO 1000. Despite the detailed metadata carried in these data formats, if they use the literal SI symbols, the unit expressions can be ambiguous, which could be hazardous for these enterprises. A UN [49] and a private [50] organisation have developed recommendations for unambiguous unit codes. The latter uses a period as the product symbol, which is disallowed in typeset SI expressions and which is thus bound to cause confusion to programmers and users. OASIS is developing a general mark-up language for units (UnitsML) [51], for embedding in other XML formats. UnitsML represents the concepts of quantity, dimension, counted-entity, unit and prefix for different languages, unit systems and character sets, but does not specify the symbols for units, operators and exponents, thus deferring this problem.

### *Unit ambiguity in typeset text*

While SI quantity expressions written in typeset text (papers, web pages, books, magazines and newspapers) are intended for human readers, increasingly they are being read, parsed and analysed by software being developed for the Semantic Web [52]. The syntactic ambiguity of the SI will only make this task harder and the results poorer.

#### Precedence in unit parsing

Unit-aware calculators [39, 40] typically implement prefix-precedence and can correctly interpret quantity algebra like '3.2 L/ha*40 km^2'.

However, there is a class of unit expressions that contain scaling factors rather than prefixes. A common example is the unit 'L/100 km', which does not fit the SI definition of a derived unit, but is in widespread legal use in Europe and Australia to express automotive fuel consumption. Calculators that use standard mathematical precedence [40] to parse expressions containing such units (e.g. '0.1 gal/mile in L/100 km') yield meaningless results.

#### Writing a quantity expression

The long absence of a SI rule for writing a quantity expression has resulted in poor media and general SI usage. Many publications omit the number-product symbol, e.g. '300 km/h winds'.

Note that for all versions of the SI, the space character in a quantity expression (e.g. '2.997 924 58$\times$10$^8$ m s$^{-1}$') has multiple meanings, representing the digit padding, scientific-notation padding, number–unit product symbol and unit–unit product symbol. This presents some challenges for parsing.

#### Inability to write, or at least inconvenience in writing, correct SI expressions

The absence of keyboard characters for the SI symbols dot operator, multiplication sign, degree, arcminute, arcsecond, mu and omega creates unnecessary difficulty for authors, who must generate these characters via system software, by key combinations or the use of special 'insert' functions in application software. Even in writing the simple quantity expression "1.234$\times$10$^5$ N·m", there are no keyboard characters for two of the three multiplication symbols.

## Steps towards a machine- and user-friendly SI notation

Users and interpreters of the SI, especially computer programmers who are not necessarily knowledgeable about unit systems, rely on the specifications of the SI published by the BIPM and ISO, and so the specifications as well as the SI units should be clear, unambiguous and harmonised. The development and acceptance of mark-up languages and codesets is slow, partly because this activity is largely voluntary and international. Regardless of whether these tools are completed and adopted, the representation of SI units at least can be made unambiguous with some simple changes.

The SI should not stand on some higher plane, ignoring the grubby reality of machine implementation. Suggestions for change, matching the principles outlined in section 2, are made below.

#### The characters for the SI symbols should be specified

If the SI is to be communicated to machines as well as humans, then its characters simply have to be uniquely identified. This is not straightforward, as there are many candidate characters in different characters sets that could be mapped to the SI symbols. Clearly, the legacy ASCII character set is too limited and the ubiquitous ISO-8859-1 has omissions. Thus, SI characters should be chosen from the Unicode character set, with mu and omega, in particular, being identified as Greek letters rather than the special Unicode symbols 'micro sign' and 'Ohm sign', which are provided for backward compatibility.

#### The SI should specify a notation for plain-text quantity and unit expressions

Plain text is widely used in computer applications, and it is time there was a standardised way of representing SI quantity and unit expressions.

### The SI notation should be designed to generate unambiguous unit expressions

The SI notation (including the 'units accepted for use with the SI') can be made unambiguous by these simple changes [53]:

- Replace the prefix 'da' with 'D'
- Change the symbol for 'day' from 'd' to 'day'
- Require an explicit product symbol in derived units

### The SI specifications should explicitly list the precedence rules for parsing unit expressions

Unit expressions like 'L/100 km' can be parsed correctly by adopting an additional precedence rule that binds the number to the unit with higher priority than the division operator. In effect, this rule adds parenthesis around the factor and unit, i.e. L/(100 km). Note that NIST [54] already writes this unit as "L/(100 km)".

Thus, the precedence order for parsing a unit expression should be as follows:

1. Prefix-unit binding
2. Parentheses
3. Exponentiation
4. Factor-unit binding
5. Multiplication and division

### The SI specifications should maintain the rules for forming quantity expressions

Both SI-8 and ISO 80000, the current versions of the BIPM and ISO standards for the SI, explicitly specify how to form a quantity expression. These specifications should be maintained and be recasted in the direct voice, e.g. "a quantity expression is formed by a numerical expression followed by a space and a unit, the space representing a multiplication sign…."

### The SI symbols should use Latin and keyboard characters, where possible

Once SI unit expressions can be written unambiguously, it would be useful to address their ease-of-use, as this will affect their rate of correct usage, just as the convenience of a unit itself will affect its rate of adoption [55]. For human convenience:

- Replace mu and omega ($\mu$, $\Omega$) with the Latin 'u' and 'Ohm'; these symbols have a precedent as they were recommended in ISO 2599.
- Replace the dot operator with the asterisk (*) as a unit–unit multiplication symbol. This has the advantages that it is a keyboard character, is more visible than a dot operator and is widely known as a product symbol on keyboards and calculators. It has an additional subtle advantage that it highlights that the nature of a metrological product (e.g. N*m) is different from an algebraic product represented by a space or multiplication sign (e.g. '30 m', '$4.0\times10^7$ m').

## A route to change?

How are ongoing changes in the SI notation, as distinct from its metrological definitions, to be assessed and standardised? It is possible that the BIPM may not be the best organisation to manage this.

The BIPM was mandated in 1875 by international diplomatic treaty to develop the standards, artefacts and realisations for the world's metric units and to coordinate the NMI's in world metrology. By default, it also took responsibility for the notation of those units, which it did from the scientific viewpoint. The BIPM's current description of the SI, including commentary and history, runs to 180 pages and cannot be described as a 'standard'.

ISO has extensive experience in balancing technical, industrial, government and consumer interests in the development of international standards. It already publishes standards in related areas such as nomenclature, symbols, quantities, character sets and data transmission, so is ideally placed to manage the SI notation.

It seems logical that the BIPM should be the scientific agency which defines and maintains the SI units, and that the specification of the SI notation should be the responsibility of ISO.

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